Vectors: Components and Operations

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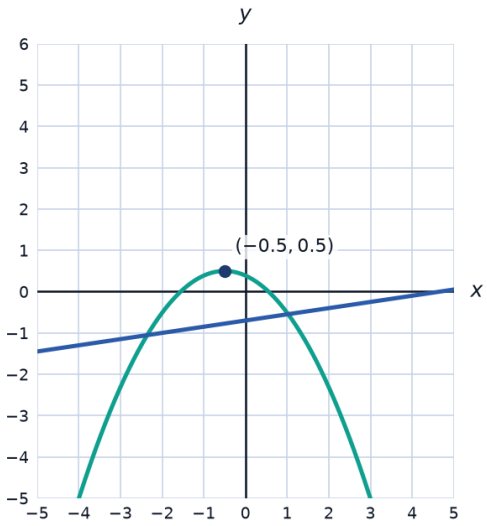
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Vector arithmetic

Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

2 [Foundational] • Target: Vector arithmetic

Write the vector of magnitude 12 making an angle of $150°$ with the positive x-axis in component form (exact values).



3 [Foundational] • Target: Applications

A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

b Find the direction of the resultant, as an angle north of east to two decimal places.

4 [Foundational] • Target: Applications

Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

5 [Foundational] • Target: Applications

A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places.

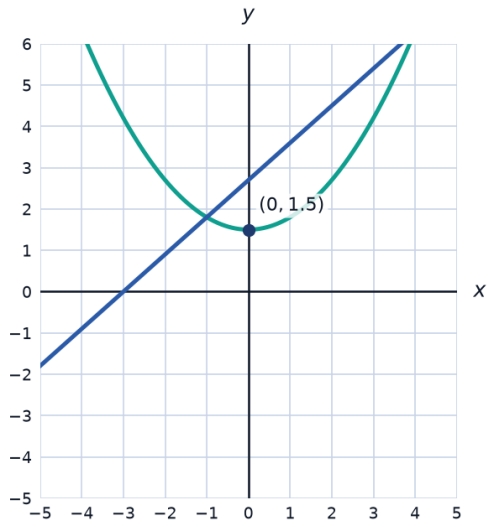
b If the river is 200 m wide, find how long the crossing takes.

c Find how far downstream (east) the boat lands.

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6 [Foundational] • Target: Position vectors and geometry

Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.



7 [Foundational] • Target: Position vectors and geometry

Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Position vectors and geometry

Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

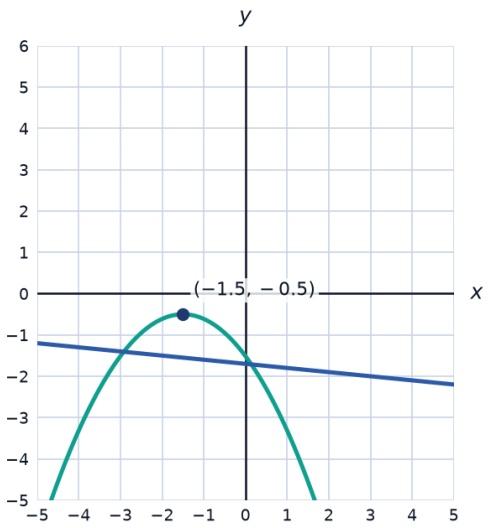
9 [Intermediate] • Target: More applications

An airplane has an airspeed of 305 mph and flies on a heading $7.5°$ west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).

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10 [Intermediate] • Target: More applications

Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.



11 [Intermediate] • Target: Finding direction angles

Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.

12 [Intermediate] • Target: Finding direction angles

A vector has magnitude 10 and direction angle $200°$. Write it in component form, rounded to two decimal places.

13 [Intermediate] • Target: Vector equations and unknowns

Find scalars c_1 and c_2 such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.

14 [Intermediate] • Target: Parallel and equal vectors

Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

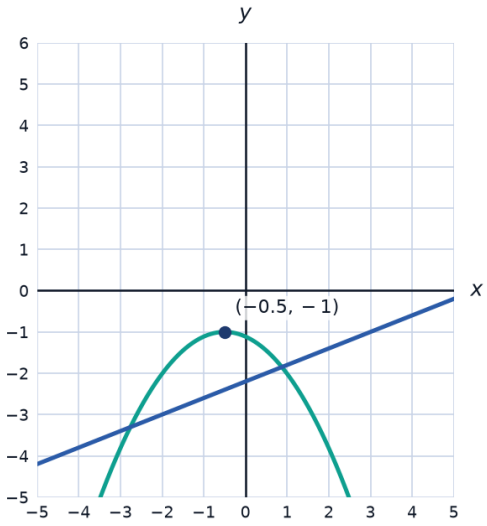
15 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 1.01 Functions and Function Notation

Find the value of k so that $\langle k, 6 \rangle$ is parallel to $\langle 4, 8 \rangle$.

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16 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 1.02 Domain and Range

A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.



17 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 1.03 Average Rate of Change

Three ropes pull on a ring with forces $\langle 10, 0\rangle$, $\langle -5, 8.66\rangle$, and $\langle -5, -8.66\rangle$ N. Find the net force, and explain what this implies physically.

18 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350\rangle$ N and the other exerts $\langle 200, 400\rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.

19 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.02 Radian Measure and Arc Length

A swimmer aims to swim directly across a river at 1.5 m/s relative to the water, while the current flows at 0.8 m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.

20 [Challenging] • Target: More applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Given vectors $\mathbf{u} = \langle 4, -3\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vector arithmetic

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

22 [Foundational] • Target: Applications

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

a. 7

b. $\sqrt{31}$

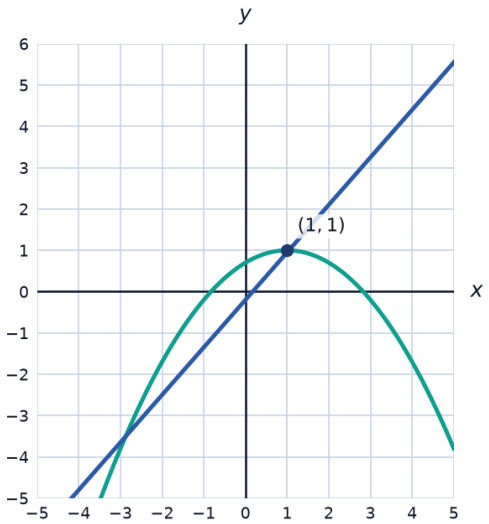
c. $\sqrt{69}$

d. 49

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23 [Foundational] • Target: Position vectors and geometry

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.



a. 44

b. 45

c. 50

d. 54

24 [Foundational] • Target: More applications

In ΔABC, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides ΔABC into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

a. $2\sqrt{6}$

b. $\frac{7\sqrt{2}}{2}$

c. $4\sqrt{3}$

d. $6\sqrt{2}$

25 [Foundational] • Target: Finding direction angles

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

a. 34

b. 64

c. 80

d. 128

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26 [Foundational] • Target: Vector equations and unknowns

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

27 [Foundational] • Target: Parallel and equal vectors

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2

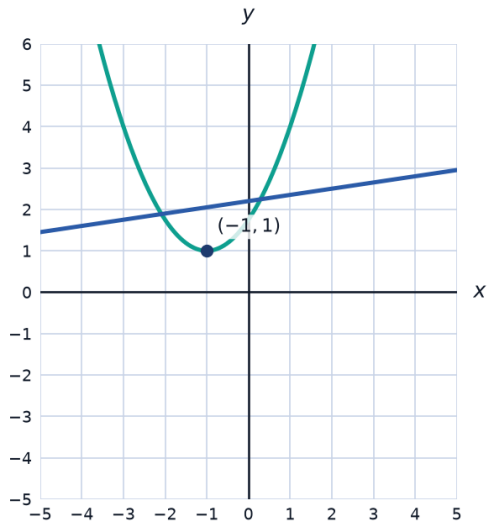
b. 2

c. -4

d. 4

28 [Foundational] • Target: Vectors in three real-world scenarios

In a championship playoff, two teams, Team A and Team B, play a best-of-five series, where the first team to win three games wins the series. The games are independent, cannot end in a tie, and in each individual game, the probability that Team A wins is p, where $0 < p < 1$ and $p \neq \frac{1}{2}$. Suppose the probability that the series is decided in exactly 3 games is equal to the probability that the series requires a decisive 5th game. What is the value of the product $p(1-p)$?



a. $\frac{\sqrt{33}-3}{12}$

b. $\frac{\sqrt{33}+3}{12}$

c. $\frac{\sqrt{33}-3}{6}$

d. $\frac{6-\sqrt{33}}{12}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Vector arithmetic**

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32

b. -1

c. $\frac{32}{3}$

d. 16

30 **[Intermediate]** • **Target: Applications**

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the x-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose x-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of k?

a. 400

b. 225

c. 256

d. 625

31 **[Intermediate]** • **Target: Position vectors and geometry**

A vector $\mathbf{v}$ is defined by its component form $\langle 2a + 1, 3 - b \rangle$. This vector is then transformed by a translation represented by the vector $\mathbf{t} = \langle 5, -2 \rangle$. The resulting transformed vector, $\mathbf{v}'$, is given as $\langle 12, 1 \rangle$. Determine the values of the parameters a and b.

a. a = 3, b = -4

b. a = 4, b = 3

c. a = 3, b = 4

d. a = 4, b = -3

32 **[Intermediate]** • **Target: More applications**

A solid cone has a height of 24 cm. A smaller cone is cut from the top of the larger cone by a plane parallel to the base. The smaller cone has a height of 8 cm. If the radius of the base of the larger cone is 12 cm, what is the radius of the base of the smaller cone?

a. 4 cm

b. 6 cm

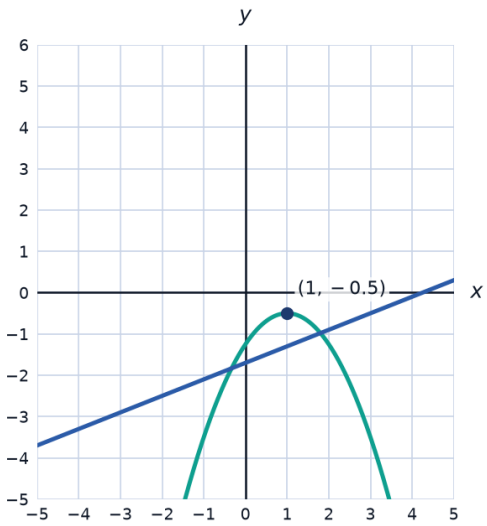
c. 8 cm

d. 5 cm

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33 [Intermediate] • Target: Finding direction angles

The polynomial $P(x) = x^4 - 8x^3 + ax^2 + bx + c$ has four real roots that form an arithmetic progression. If the sum of the squares of the four roots is 36, what is the value of the constant term c?



a. -15

b. -9

c. 9

d. 15

34 [Intermediate] • Target: Vector equations and unknowns

A cone has a height of 18 cm. A smaller cone is formed by slicing the top of the original cone with a plane parallel to the base. The smaller cone is similar to the original cone, and its height is 12 cm. If the radius of the base of the original cone is 9 cm, what is the radius of the base of the smaller cone?

a. 6 cm

b. 5 cm

c. 4.5 cm

d. 3 cm

35 [Intermediate] • Target: Parallel and equal vectors

A right circular cone has a height of 12 cm. A smaller cone is formed by slicing the top of the larger cone with a plane parallel to the base. The smaller cone is similar to the larger cone, and its height is 6 cm. If the radius of the base of the larger cone is 9 cm, what is the radius of the base of the smaller cone?

a. 4.5 cm

b. 5 cm

c. 6 cm

d. 7.5 cm

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36 [Intermediate] • Target: Vectors in three real-world scenarios

In the diagram below, lines DE and BC are parallel. If AD = 6, DB = 4, and AE = 9, what is the length of EC?

a. 6

b. 7

c. 8

d. 10

37 [Intermediate] • Target: Vector arithmetic

A vector $\mathbf{v}$ is defined by its initial point $P = (-5, 3)$ and its terminal point $Q = (4, -2)$. What is the component form of vector $\mathbf{v}$?

a. $\langle -9, 5 \rangle$

b. $\langle 9, -5 \rangle$

c. $\langle -1, 1 \rangle$

d. $\langle 1, -1 \rangle$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. Find the component form of $\mathbf{v}$.

a. $\langle 3, 4 \rangle$

b. $\langle 7, 10 \rangle$

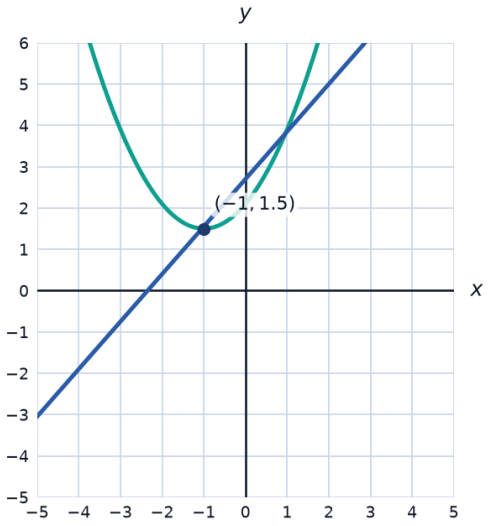
c. $\langle -3, -4 \rangle$

d. $\langle 4, 3 \rangle$

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39 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 2.02 Radian Measure and Arc Length

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?



a. $\langle -5, 0 \rangle$

b. $\langle 5, 0 \rangle$

c. $\langle -3, 0 \rangle$

d. $\langle 3, 0 \rangle$

40 [Challenging] • Target: More applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Consider the transformation grid below. If the original point A is mapped to A', what is the vector representing this translation?

a. $\langle -5, 0 \rangle$

b. $\langle -6, 0 \rangle$

c. $\langle 5, 0 \rangle$

d. $\langle 6, 0 \rangle$

41 [Challenging] • Target: Finding direction angles • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider the vector $\mathbf{v}$ directed from the point $P = (2, 5)$ to the point $Q = (7, 1)$. What is the component form of $\mathbf{v}$?

a. $\langle -5, -4 \rangle$

b. $\langle 5, 4 \rangle$

c. $\langle 5, -4 \rangle$

d. $\langle -5, 4 \rangle$

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42 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

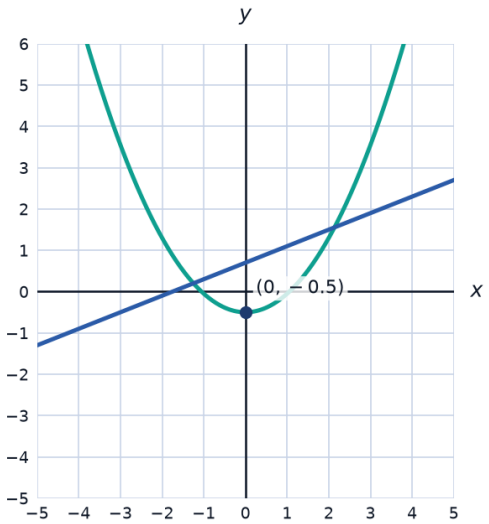
b. 20

c. 25

d. 22

43 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the real number(s) equal to $(-32)^{3/5}$.



a. 8

b. Undefined

c. -8

d. -32^3

44 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 3.03 Sum and Difference Formulas

What are the equations of the asymptotes of the hyperbola $\frac{(x-1)^2}{9} - \frac{(y+2)^2}{4} = 1$?

a. $y + 2 = \pm\frac{2}{3}(x - 1)$

b. $y + 2 = \pm\frac{3}{2}(x - 1)$

c. $y = \pm\frac{2}{3}x$

d. $y - 2 = \pm\frac{2}{3}(x + 1)$

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45 [Challenging] • Target: Vector arithmetic • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

a. 70

b. 64

c. 62

d. 65

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